

A Quantitative Study of Technical Analysis Indicators and Intraday Trading Strategies on Indian Equity Market Data

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April 2026



Outline

- 1 Introduction
- 2 Background and Technical Indicators
- 3 Data Description and Preprocessing
- 4 Signal Analysis
- 5 Single-Asset Backtesting
- 6 Cross-Sectional Backtesting
- 7 Conclusion and Future Work

Motivation

- Intraday markets provide dense 5-minute data and enable practical algorithmic execution.
- Technical indicators are widely used but often applied heuristically.
- In liquid equities, small statistical edges are fragile due to costs and microstructure noise.
- Therefore, indicator effectiveness must be **empirically validated**, not assumed.

Problem Statement

Core Question

Do widely used technical indicators exhibit predictive power at 5-minute horizons, and can this be translated into stable trading performance in the NIFTY50 universe?

Objectives

- Quantify predictive power using event-based forward returns.
- Evaluate robustness across assets and time horizons.
- Design and backtest intraday strategies under realistic constraints.

Contributions

- Unified pipeline linking signal-level statistics to trading performance.
- Cross-asset pooled analysis with multiple-testing control.
- Comparative evaluation of single-asset and multi-asset strategies.

Technical Analysis: Overview

- Technical analysis studies price and volume to identify patterns and predict short-term movements.
- Core assumption: **information is reflected in prices** and patterns may repeat.
- Indicators act as **quantitative summaries** of:
 - Trend
 - Momentum
 - Volatility
 - Volume
- Goal in this work: evaluate whether these indicators contain **predictive signal** in intraday data.

Notation (Key Variables)

- P_t : price at time t , V_t : volume
- n : lookback window
- $r_t = \frac{P_t - P_{t-1}}{P_{t-1}}$: return
- SMA_t , EMA_t : moving averages
- σ_t : rolling volatility
- H_t, L_t : high and low, C_{t-1} : previous close

Purpose: Standardize notation for indicator definitions.

Trend Indicators

Simple Moving Average (SMA):

$$SMA_t = \frac{1}{n} \sum_{i=0}^{n-1} P_{t-i}$$

Exponential Moving Average (EMA):

$$EMA_t = \alpha P_t + (1 - \alpha) EMA_{t-1}$$

MACD:

$$MACD_t = EMA_t^{(12)} - EMA_t^{(26)}$$

- SMA: smoother, slower response
- EMA: faster, more reactive
- MACD: combines trend + momentum via crossovers

Momentum Indicators

Relative Strength Index (RSI):

$$RSI = 100 - \frac{100}{1 + RS}$$

Stochastic Oscillator:

$$K_t = \frac{P_t - L_n}{H_n - L_n} \times 100$$

Rate of Change (ROC):

$$ROC = \frac{P_t - P_{t-n}}{P_{t-n}} \times 100$$

- Capture **speed and extremeness** of price moves
- Identify **overbought / oversold** conditions

Volatility Indicators

Bollinger Bands:

$$Upper = SMA_t + k\sigma_t, \quad Lower = SMA_t - k\sigma_t$$

True Range (ATR component):

$$TR_t = \max(H_t - L_t, |H_t - C_{t-1}|, |L_t - C_{t-1}|)$$

Rolling Volatility:

$$\sigma_t = \sqrt{\frac{1}{n} \sum (P_{t-i} - \mu)^2}$$

- Measure **risk and dispersion**
- Provide **regime context** (high vs low volatility)

Volume Indicators

On-Balance Volume (OBV):

$$OBV_t = OBV_{t-1} + \begin{cases} V_t & P_t > P_{t-1} \\ -V_t & P_t < P_{t-1} \\ 0 & \text{otherwise} \end{cases}$$

VWAP:

$$VWAP_t = \frac{\sum P_t V_t}{\sum V_t}$$

Accumulation/Distribution:

$$AD_t = AD_{t-1} + MF_t \cdot V_t$$

- Capture **market participation**
- Identify **buying/selling pressure**

Indicator Families Summary

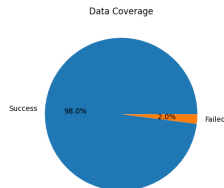
Indicator	Category	Primary Use
SMA, EMA, MACD	Trend	Direction, crossovers
RSI, Stochastic, ROC	Momentum	Speed, extremes
Bollinger, ATR, σ_t	Volatility	Risk, regimes
OBV, VWAP, A/D	Volume	Participation, pressure

Why These Indicators?

- Widely used in practice and literature
- Computationally efficient for intraday data
- Interpretable signals (trend, reversal, volatility shifts)
- Suitable for testing:
 - Predictive power
 - Strategy construction
 - Signal filtering

Data Source

- 5-minute intraday OHLC data from Yahoo Finance.
- NIFTY50 universe: 49 assets successfully retrieved (1 missing).
- Time period: Jan 2026 – Apr 2026 (~60 trading days).
- All timestamps aligned to **IST** within market hours (09:15–15:30).



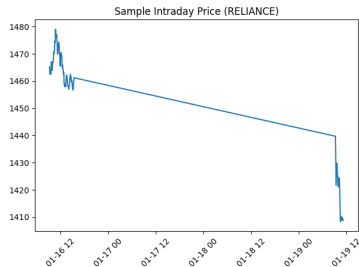
Data Preprocessing

- Chronological sorting to preserve time structure.
- Removal of duplicate timestamps.
- Missing values handled via forward fill.
- Validation to ensure consistent dataset length and format.

Outcome: Clean, aligned time series suitable for indicator computation.

Intraday Structure

- Each trading day represented as sequential 5-minute intervals.
- Non-trading hours excluded.
- Ensures all observations correspond to active market periods.



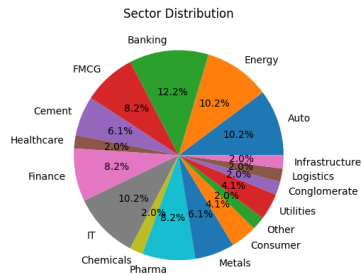
Data Representation

- Time-indexed structure: each row = one 5-minute interval.
- Columns: Open, High, Low, Close (OHLC).
- Enables:
 - Rolling computations
 - Indicator construction
 - Proper temporal alignment

Dataset Overview

Metric	Value
Total Assets Considered	50
Successfully Retrieved	49
Failed Retrievals	1
Sampling Frequency	5-minute intervals
Observation Window	~60 trading days
Time Period	Jan 2026 – Apr 2026
Price Fields	Open, High, Low, Close

Sector Coverage



Objective

Core Question

Given an indicator signal, does it exhibit predictive power over future returns?

- Evaluate statistical and economic significance
- Focus on:
 - Directional consistency
 - Horizon persistence
 - Cross-asset stability

Methodology Overview

- Event-based framework (no trade simulation)
- Signals generated per asset
- Forward returns computed post-signal
- Pooled across assets for market-level inference

Event-Based Signals

- Record only **state transitions** (false $\rightarrow$ true)
- Avoid repeated counting in persistent conditions
- Each signal = distinct market event

Benefit: Unbiased sampling and cleaner inference

Forward Returns

$$r_{t,h} = \frac{P_{t+h} - P_t}{P_t}$$

- Horizons: $h \in \{1, 5, 10\}$
- Captures immediate to short-term effects

Statistical Evaluation

- Metrics: mean, win rate, standard deviation
- Hypothesis:

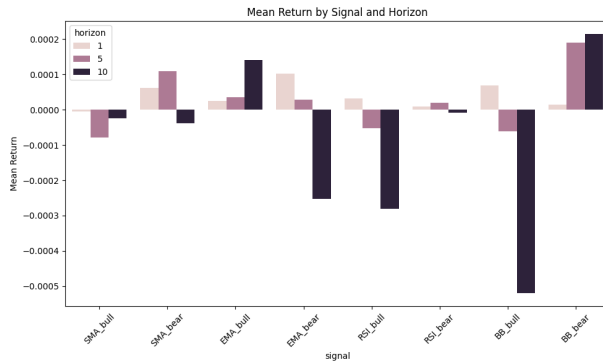
$$H_0 : \mathbb{E}[r_{t,h}] = 0$$

- Multiple testing: **FDR correction**

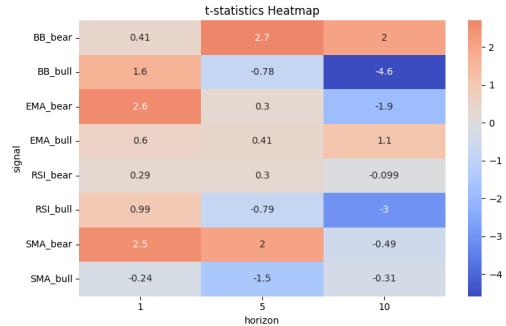
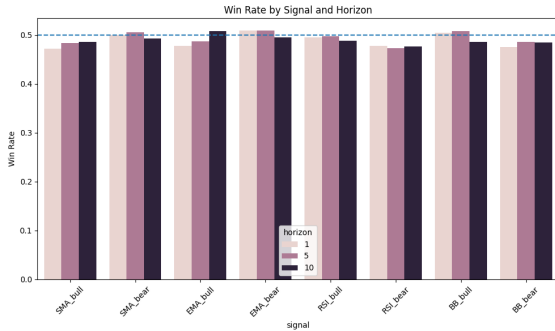
Indicators and Setup

- Indicators:
 - Trend: SMA, EMA
 - Momentum: RSI
 - Volatility: Bollinger Bands
- Dataset:
 - 49 NIFTY50 stocks, ~60 days
 - 3.5k–8.3k events per setup

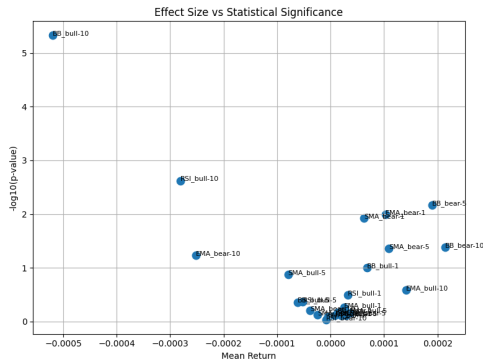
Pooled Signal Performance



Win Rates and Significance



Effect Size vs Significance



Key Findings

- Mean returns ≈ 0
- Win rates $\approx 50\%$
- No consistent horizon effects
- No signal survives FDR correction

Statistical vs Economic Significance

- Large sample $\Rightarrow$ high statistical power
- Small effects may appear significant
- But:
 - Magnitude $\ll$ transaction costs
 - No practical tradability

Conclusion

- No robust predictive power from standalone indicators
- Effects are weak, unstable, and economically negligible
- Consistent with semi-strong market efficiency
- Motivates strategy design and filtering

Objective

Core Question

Which intraday strategies deliver robust, cost-aware performance on 5-minute data?

- Evaluate baseline strategies under realistic costs
- Test filtering and combinations for robustness
- Analyze per-asset and cross-asset performance

Strategy Design

Baselines

- Random (benchmark)
- SMA crossover
- RSI mean reversion
- VWAP reversion
- ORB

Filtered / Combined

- ORB_VWAP (confirmation)
- RSI_TREND (trend filter)
- SMA_VOL (volatility filter)

Constraints

- Long-only, intraday (EOD exit)

Backtesting Framework

- Event-driven execution (no repeated signals)
- Execution at next candle open (no look-ahead)
- Full capital allocation per trade
- Intraday-only (no overnight exposure)

Costs and Risk Controls

Transaction Costs

- Fee: 0.05%
- Slippage: 0.02%

Risk Controls

- Stop-loss: 1%
- No take-profit (avoid overfitting)

Evaluation Metrics

Performance

- Total return
- Sharpe, Sortino
- Max drawdown

Trade-Level

- Win rate
- Profit factor
- Expectancy

Ranking

- Composite (return + Sharpe + drawdown)

Global Results (All Assets)

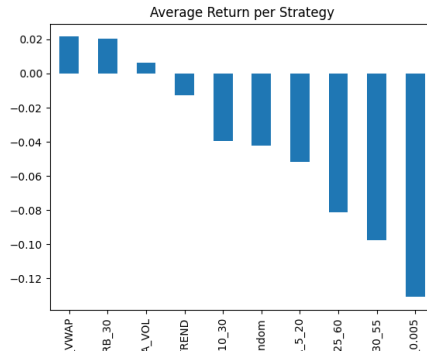


Figure: Average return by strategy across assets.

Risk–Return Landscape

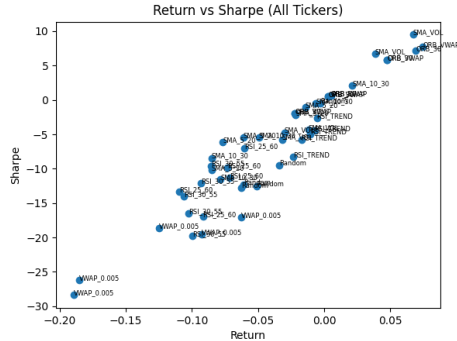
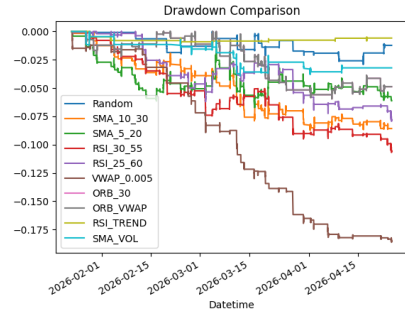
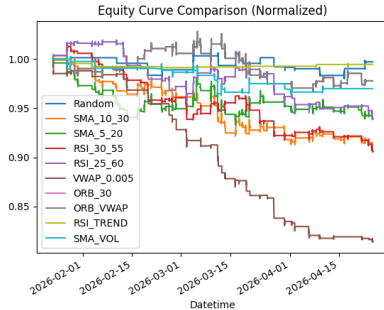


Figure: Return vs. Sharpe across strategies.

Case Study: RELIANCE_NS



Key Findings

- Most standalone strategies fail after costs
- ORB is the strongest baseline signal
- VWAP is effective as a **confirmation filter**
- Combining signals reduces noise and improves stability
- Strong cross-asset variation $\Rightarrow$ robustness requires aggregation

Conclusion

- Simple indicators are insufficient for profitability
- Filtering improves structure but not consistency
- Motivates transition to **multi-asset analysis**

Cross-Sectional Setup

- 49 NIFTY50 stocks evaluated independently under identical rules
- No portfolio construction; aggregation used to assess robustness
- Trade-level metrics isolate intrinsic signal expectancy

Pure Indicator Strategies



Figure: All strategies lie in the negative-return region ($PF < 1$, $Sharpe < 0$).

Filtered Strategies

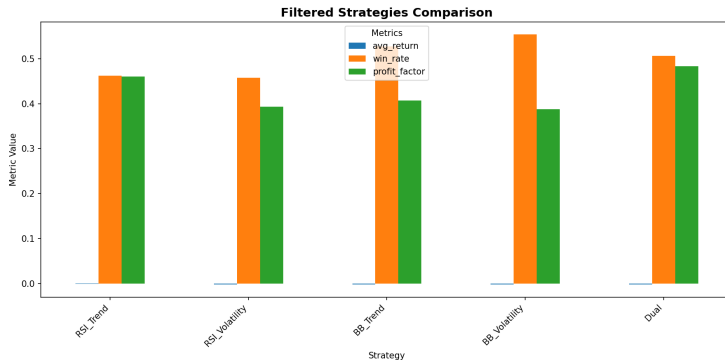


Figure: Filtering reduces trades and raises win rates, but expectancy remains negative.

Combined Strategies

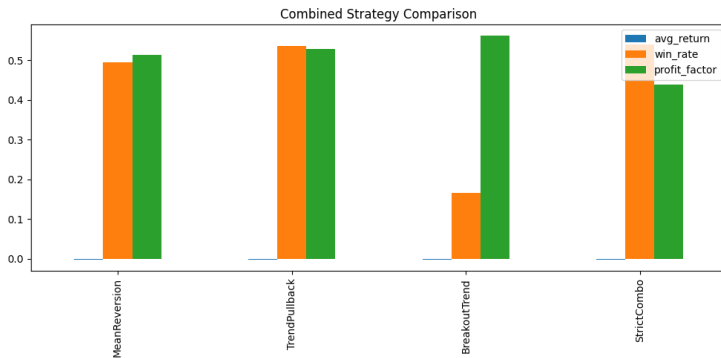
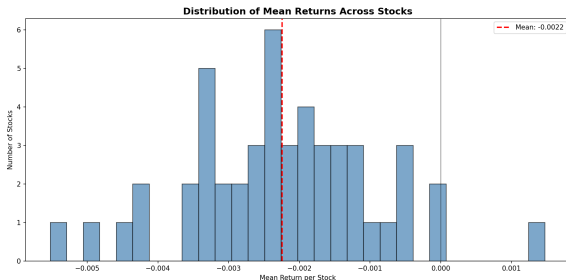
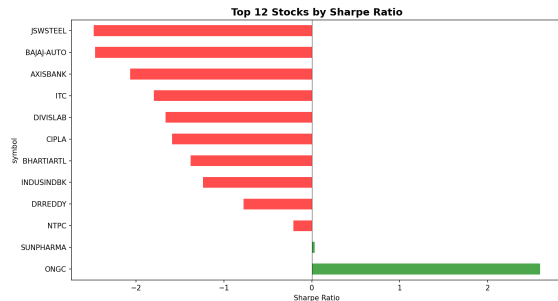


Figure: Improved structure (confirmation, payoff asymmetry) but still unprofitable.

Stock-Level Robustness



- Only ~4% of stocks are profitable
- Cross-sectional distribution is centered below zero



Synthesis

- **Win-rate paradox:** $p \approx 40\text{--}55\%$, but losses dominate gains
- **Cost drag:** 0.07% per trade overwhelms weak signals
- **Execution lag:** Next-bar execution degrades short-horizon edge
- **Cross-sectional instability:** Performance does not generalize

Conclusion

- Signal-level predictability exists but is economically negligible after FDR control.
- Strategies show negative returns, $PF < 1$, and Sharpe < 0 after costs.
- ORB + VWAP improves structure, but no consistent cross-asset profitability.
- Win-rate paradox: $p > 50\%$ but losses dominate gains.

Future Work

- Regime detection and volatility conditioning.
- Execution-aware modeling (latency, slippage, partial fills).
- Portfolio construction and capital allocation.
- Dynamic position sizing and longer regime-diverse datasets.

Thank you